The origin of ferromagnetism and the Hubbard model

part 2: exchange interaction in two-electron systems

Advanced Topics in Statistical Physics
by Hal Tasaki

§ symmetry of wave function and spin states

nonrelativistic description of a single electron

position $\mathbf{r} \in \mathbb{R}^3$, spin $\sigma \in \{\uparrow, \downarrow\}$

wave function $\varphi(\mathbf{r}, \sigma)$

corresponding ket state

$$|\varphi\rangle = \sum_{\sigma=\uparrow,\downarrow} \int d^3\mathbf{r}\, \varphi(\mathbf{r}, \sigma)\, |\mathbf{r}\rangle\, |\sigma\rangle \quad (1)$$

two electrons $\mathbf{r}_1, \mathbf{r}_2 \in \mathbb{R}^3$, $\sigma_1, \sigma_2 \in \{\uparrow, \downarrow\}$

wave function $\Phi(\underbrace{\mathbf{r}_1, \sigma_1}_{\text{electron A}}; \underbrace{\mathbf{r}_2, \sigma_2}_{\text{electron B}})$

electrons are fermions

$\boxed{\text{the wave function picks up a factor of } -1 \text{ when we swap the labels of two electrons !}}$

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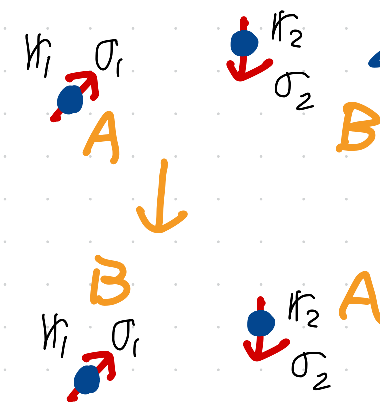


$$\underset{A}{\underbrace{\Phi(r_1,\sigma_1}};\underset{B}{\underbrace{r_2,\sigma_2}})=-\Phi(\underset{A}{\underbrace{r_2,\sigma_2}};\underset{B}{\underbrace{r_1,\sigma_1}}) \quad (1)$$

NO PHYSICAL OPERATIONS ON ELECTRONS!!

we simply "rename" them

Symmetry of (coordinate-dependent) wave functions

① $\sigma_1=\sigma_2=\uparrow$

$$\Psi_{\uparrow\uparrow}(r_1,r_2):=\Phi(r_1,\uparrow;r_2,\uparrow) \quad (2)$$

$$(1)\Rightarrow \Psi_{\uparrow\uparrow}(r_1,r_2)=-\Psi_{\uparrow\uparrow}(r_2,r_1) \quad (3)$$

familiar antisymmetric wave functions

similarly

② $\sigma_1=\sigma_2=\downarrow$

$$\Psi_{\downarrow\downarrow}(r_1,r_2):=\Phi(r_1,\downarrow;r_2,\downarrow) \quad (4)$$

$$(1)\Rightarrow \Psi_{\downarrow\downarrow}(r_1,r_2)=-\Psi_{\downarrow\downarrow}(r_2,r_1) \quad (5)$$

▷ case with $\sigma_1 + \sigma_2 = 0$

$$\Psi_{trip}(r_1, r_2) := \Phi(r_1, \uparrow; r_2, \downarrow) + \Phi(r_1, \downarrow; r_2, \uparrow)$$

$$\Psi_{trip}(r_1, r_2) \overset{p2-(1)}{=} -\Phi(r_2, \downarrow; r_1, \uparrow) - \Phi(r_2, \uparrow; r_1, \downarrow)$$

$$= -\{\Phi(r_2, \uparrow; r_1, \downarrow) + \Phi(r_2, \downarrow; r_1, \uparrow)\}$$

$$= -\Psi_{trip}(r_2, r_1) \quad (1)$$

antisymmetric

$$\Psi_{sing}(r_1, r_2) := \Phi(r_1, \uparrow; r_2, \downarrow) - \Phi(r_1, \downarrow; r_2, \uparrow)$$

$$\Psi_{sing}(r_1, r_2) \overset{p2-(1)}{=} -\Phi(r_2, \downarrow; r_1, \uparrow) + \Phi(r_2, \uparrow; r_1, \downarrow)$$

$$= \Phi(r_2, \uparrow; r_1, \downarrow) - \Phi(r_2, \downarrow; r_1, \uparrow)$$

$$= \Psi_{sing}(r_2, r_1) \quad (2)$$

Symmetric!

they are not bosons.

corresponding spin states

spin swap operator $\hat{\mathcal{S}}$: $(\hat{\mathcal{S}}\Psi)(r_1,\sigma_1;r_2,\sigma_2) = \Psi(r_1,\sigma_2;r_2,\sigma_1)$ (1)

obviously $\hat{\mathcal{S}}\Psi_{\uparrow\uparrow} = \Psi_{\uparrow\uparrow}$ (2)

$\hat{\mathcal{S}}\Psi_{\downarrow\downarrow} = \Psi_{\downarrow\downarrow}$ (3)

and $\hat{\mathcal{S}}\Psi_{trip} = \Psi_{trip}$ (4)

spin triplet $\boxed{S_{tot} = 1}$

$$\begin{cases} |\uparrow\rangle_1|\uparrow\rangle_2 \\ \frac{1}{\sqrt{2}}\left(|\uparrow\rangle_1|\downarrow\rangle_2 + |\downarrow\rangle_1|\uparrow\rangle_2\right) \\ |\downarrow\rangle_1|\downarrow\rangle_2 \end{cases}$$

on the other hand

$\hat{\mathcal{S}}\Psi_{sing} = -\Psi_{sing}$ (5)

spin singlet $\boxed{S_{tot} = 0}$

$$\frac{1}{\sqrt{2}}\left(|\uparrow\rangle_1|\downarrow\rangle_2 - |\downarrow\rangle_1|\uparrow\rangle_2\right)$$

$$|\Phi_{12}\rangle = |\Psi_{12}\rangle \otimes \overbrace{|\Xi_{12}\rangle}^{\text{spin part}} \quad (6)$$

$\downarrow$

antisymmetric $\begin{cases} \text{antisymmetric} \quad \text{symmetric} \leftarrow (2),(3),(4) \\ \text{symmetric} \quad \text{antisymmetric} \leftarrow (5). \end{cases}$

§ Heisenberg's exchange interaction

no magnetic field
no LS coupling

spin-independent Hamiltonian for two electrons

$$\hat{H} = \frac{|\hat{\mathbf{p}}_1|^2}{2m} + \frac{|\hat{\mathbf{p}}_2|^2}{2m} + V(\hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2) \quad (1)$$

$$V(\mathbf{r}_1, \mathbf{r}_2) = V(\mathbf{r}_2, \mathbf{r}_1) \quad (2)$$ any potential + interaction

Schrödinger equation

$$\left\{ -\frac{\hbar^2}{2m}\Delta_1 - \frac{\hbar^2}{2m}\Delta_2 + V(\mathbf{r}_1, \mathbf{r}_2) \right\} \Phi(\mathbf{r}_1, \sigma_1 ; \mathbf{r}_2, \sigma_2) = E\, \Phi(\mathbf{r}_1, \sigma_1 ; \mathbf{r}_2, \sigma_2) \quad (3)$$

we shall investigate the ground state, espesically, its spin state

Schrödinger equation for coordinate wave function $\Psi(\mathbf{r}_1, \mathbf{r}_2)$

$$\left\{-\frac{\hbar^2}{2m}\Delta_1 - \frac{\hbar^2}{2m}\Delta_2 + V(\mathbf{r}_1, \mathbf{r}_2)\right\}\Psi(\mathbf{r}_1, \mathbf{r}_2) = E\,\Psi(\mathbf{r}_1, \mathbf{r}_2) \qquad (1)$$

invariant under swap $\mathbf{r}_1 \leftrightarrow \mathbf{r}_2$

$\Downarrow$

one can take energy eigenfunctions to be either

$\boxed{S}$ symmetric $\Psi(\mathbf{r}_1, \mathbf{r}_2) = \Psi(\mathbf{r}_2, \mathbf{r}_1)$ (2)

or

$\boxed{A}$ antisymmetric $\Psi(\mathbf{r}_1, \mathbf{r}_2) = -\Psi(\mathbf{r}_2, \mathbf{r}_1)$ (3)

$\boxed{S} \rightarrow \Psi_{sing} \rightarrow \boxed{S_{tot} = 0}$

$\boxed{A} \rightarrow \Psi_{\uparrow\uparrow}, \Psi_{trip}, \Psi_{\downarrow\downarrow} \rightarrow \boxed{S_{tot} = 1}$

E_{sym} : the lowest energy among $\boxed{S}$

E_{anti} : the lowest energy among $\boxed{A}$

exchange interaction $J_{ex} = \frac{1}{2}\{E_{sym} - E_{anti}\}$

if $J_{ex} > 0$ the ground state is in $\boxed{A}$ → $\boxed{S_{tot} = 1}$

ferromagnetic interaction

$J_{ex} < 0$ the ground state is in $\boxed{S}$ → $\boxed{S_{tot} = 0}$

antiferromagnetic interaction

the effective spin-spin interaction arises from the symmetry of fermions!

repulsive interaction favors ferromagnetism

if $\Psi(\mathbf{r}_1, \mathbf{r}_2)$ is $\boxed{A}$ then $\Psi(\mathbf{r}_1, \mathbf{r}_2) \simeq 0$ for $\mathbf{r}_1 \simeq \mathbf{r}_2$

electrons tend to avoid with each other!

if $\Psi(\mathbf{r}_1, \mathbf{r}_2)$ is $\boxed{S}$ then no such tendency

Coulomb!

when $V(\mathbf{r}_1, \mathbf{r}_2)$ contains a repulsive interaction

the energy should be lower when electrons avoid with each other

$$E_{anti} < E_{sym} \qquad (1)$$

$$J_{ex} = \frac{1}{2}(E_{sym} - E_{anti}) > 0 \qquad (2)$$

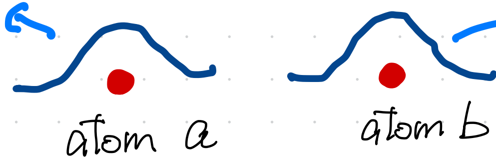
$$\boxed{\text{fermion symmetry + repulsive interaction} \rightarrow \text{ferromagnetism!!}}$$

"Heisenberg's idea"

perturbative treatment (Heisenberg 1928)

orbit around atom a

$\varphi_a(\mathbf{r})$



orbit around atom b

$\varphi_b(\mathbf{r})$

assume, for simplicity, $\langle \varphi_a | \varphi_b \rangle = 0$ (1) $\quad \|\varphi_a\| = \|\varphi_b\| = 1$ (2)

two electrons at a and b

$$\Psi_{sym}(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{\sqrt{2}} \{ \varphi_a(\mathbf{r}_1)\varphi_b(\mathbf{r}_2) + \varphi_a(\mathbf{r}_2)\varphi_b(\mathbf{r}_1) \} \quad (3)$$

$$\Psi_{anti}(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{\sqrt{2}} \{ \varphi_a(\mathbf{r}_1)\varphi_b(\mathbf{r}_2) - \varphi_a(\mathbf{r}_2)\varphi_b(\mathbf{r}_1) \} \quad (4)$$

exchange interaction ⟵ 1st order perturbation

$$J_{ex} = \frac{1}{2} \{ \langle \Psi_{sym} | \hat{V}_{int} | \Psi_{sym} \rangle - \langle \Psi_{anti} | \hat{V}_{int} | \Psi_{anti} \rangle \}$$

$$= \frac{1}{2} \int d\mathbf{r}_1 d\mathbf{r}_2 \{ \varphi_a^*(\mathbf{r}_1)\varphi_b^*(\mathbf{r}_2) V_{int}(\mathbf{r}_1, \mathbf{r}_2) \varphi_a(\mathbf{r}_2)\varphi_b(\mathbf{r}_1) + c.c. \}$$

one may have $J_{ex} > 0$ (ferromagnetism) for certain repulsive V_{int}

§ "Wigner's theorem" (appears in Lieb-Mattis 1962)

Schrödinger equation for coordinate wave function $\Psi(\mathbf{r}_1, \mathbf{r}_2)$

$$\left\{-\frac{\hbar^2}{2m}\Delta_1 - \frac{\hbar^2}{2m}\Delta_2 + V(\mathbf{r}_1, \mathbf{r}_2)\right\}\Psi(\mathbf{r}_1, \mathbf{r}_2) = E\,\Psi(\mathbf{r}_1, \mathbf{r}_2) \qquad (1)$$

theorem: for any potential $V(\mathbf{r}_1, \mathbf{r}_2) = V(\mathbf{r}_2, \mathbf{r}_1)$, the ground state of (1) is unique and symmetric, $\Psi_{GS}(\mathbf{r}_1, \mathbf{r}_2) = \Psi_{GS}(\mathbf{r}_2, \mathbf{r}_1)$. (2)

$J_{ex} < 0$ antiferromagnetic interaction is always favored in a two-electron system!

~~fermion symmetry + repulsive interaction → ferromagnetism!!~~

"Heisenberg's idea" ← NOT REALLY. he is much smarter!!

“Zur Theorie des Ferromagnetismus,” Zeit. Phys. **49** (1928), 619-636.

61

On the theory of ferromagnetism

By W. Heisenberg in Leipzig

With 1 Figure. (Received on 20 May 1928)

Translated by D. H. Delphenich

WEISS’s molecular forces will be attributed to a quantum-mechanical exchange phenomenon, and indeed, it will be treated as the exchange process that was successfully enlisted in recent times by HEITLER and LONDON in order to interpret homopolar valence forces.

he knew it does not suffice to have only 1s orbits!

The question of the sign of J_0 is much more difficult to answer. In their theory of the homopolar bond, HEITLER and LONDON make that assumption that J_0 is negative in complete generality, which would exclude ferromagnetism. For the special case in which the electrons are found to be unperturbed in the 1*s* state, it follows, in fact, from general theorems that the energy values must lie in a way that would correspond to negative values of J_0 . Such an argument is, in turn, applicable to only electrons in the 1*s* state, and one can show that J_0 will generally be positive for high principal quantum numbers. One must then deal with the expression:

$$J_0 = \frac{1}{2}\int \psi_k^\kappa \psi_k^\lambda \psi_l^\kappa \psi_l^\lambda \left(\frac{2e^2}{r_{kl}} + \frac{2e^2}{r_{\kappa\lambda}} - \frac{e^2}{r_{\kappa k}} - \frac{e^2}{r_{\lambda k}} - \frac{e^2}{r_{\lambda l}} \right) d\tau_k \, d\tau_l \,, \tag{1}$$

proof of "Wigner's theorem"

$$\tilde{\mathbf{r}} = (\mathbf{r}_1, \mathbf{r}_2) \in \mathbb{R}^6 \quad (1)$$

p10-11 $$\left\{ -\frac{\hbar^2}{2m} \Delta_{\tilde{\mathbf{r}}} + V(\tilde{\mathbf{r}}) \right\} \Psi(\tilde{\mathbf{r}}) = E\, \Psi(\tilde{\mathbf{r}}) \quad (2)$$

standard single-particle Schrödinger equation in six dimensions with "potential" $V(\tilde{\mathbf{r}})$

general theory for Sch. eq. with real potential → $\boxed{\text{appendix 2}}$

$\Downarrow$

The ground state is unique and the eigenfunction can be chosen so that $\Psi(\tilde{\mathbf{r}}) > 0$ ← $\boxed{\text{g.s. wavefunction is "nodeless"}}$

this is only possible when $\Psi(\mathbf{r}_1, \mathbf{r}_2) = \Psi(\mathbf{r}_2, \mathbf{r}_1)$ //

§ simple tight-binding model with $U=\infty$

tight-binding appoximation

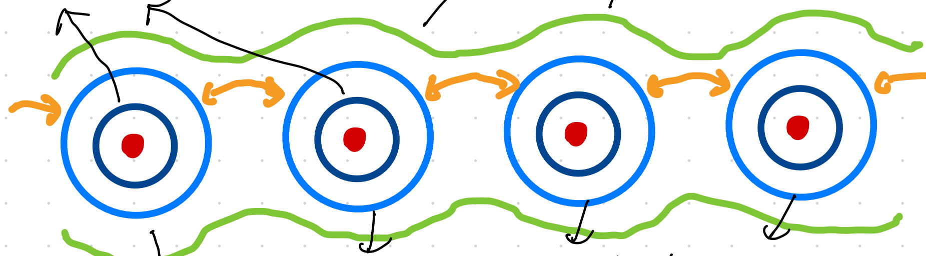
an atom

electrons
in different orbits

almost localized

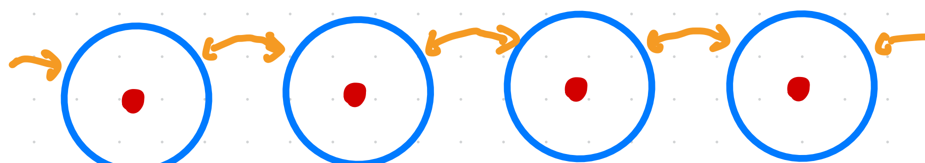
extend over the system

a crystal



mostly localized but tunnel to nearby orbits

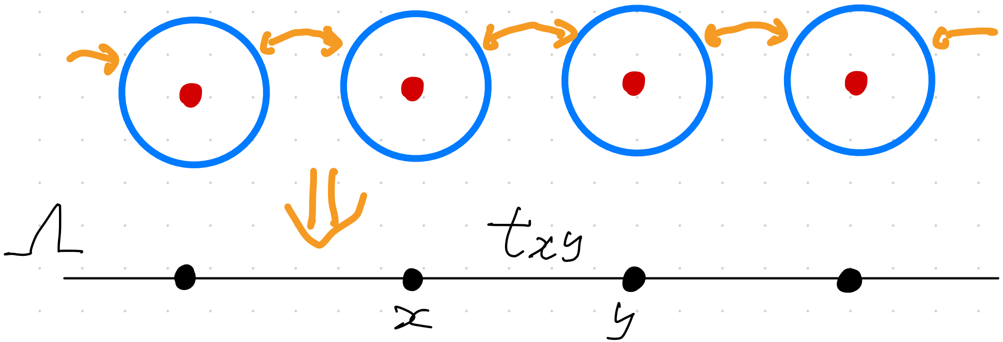
tight-binding
approximation



focus only on electrons in these orbits

▷ tight-binding model (without orbital degeneracy)

finite lattice $\Lambda \ni x, y, \dots$ lattice site $x \leftrightarrow$ orbital state around an atom



t_{xy} : hopping (tunneling) amplitude from site y to x $(x \neq y)$

(we only consider the case $t_{xy} = t_{yx} \in \mathbb{R}$)

single-particle tight-binding Schrödinger equation $\varphi_x \in \mathbb{C}$

$$E\varphi_x = \sum_{y\in\Lambda} t_{xy}\varphi_y \quad \text{for all } x \in \Lambda \qquad (1)$$

$t_{xx} (= V_x) \in \mathbb{R}$: potential at site x

simplest model in one-dimension

$$\Lambda = \{1, 2, \ldots, L\} \quad (1) \qquad \text{periodic b.c.} \quad 0 \leftrightarrow L, \quad 1 \leftrightarrow L+1$$

$$t_{xy} = \begin{cases} t \in \mathbb{R}, & |x-y| = 1 \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

Sch. eq. $E\psi_x = t\sum_{x=1}^{L}(\psi_{x+1} + \psi_{x-1})$ (3)

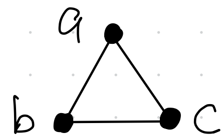
energy eigenfunction $\psi_x^{(k)} = \frac{1}{\sqrt{L}} e^{ikx}$ (4)

$$k = \frac{2\pi}{L}n, \quad (n = 1, \ldots, L) \quad (5)$$

energy eigenvalue $E_k = 2t\cos k$ (6)

iii) two-electron model on three sites

$\Lambda = \{a, b, c\}$

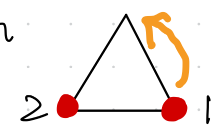


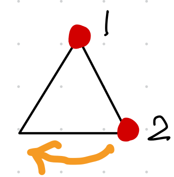
two electrons (with spin $\uparrow$ and $\downarrow$) on Λ

- an electron hops to a neighboring site with amplitude $t \in \mathbb{R}$
- strong Coulomb repulsion $\rightarrow$ two electrons cannot occupy a single site
  $\rightarrow$ $U = \infty$

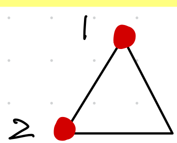
coordinate-dependent wave function $\psi(x_1, x_2)$, $x_1, x_2 \in \Lambda$, $x_1 \neq x_2$

a piece of Schrödinger equation





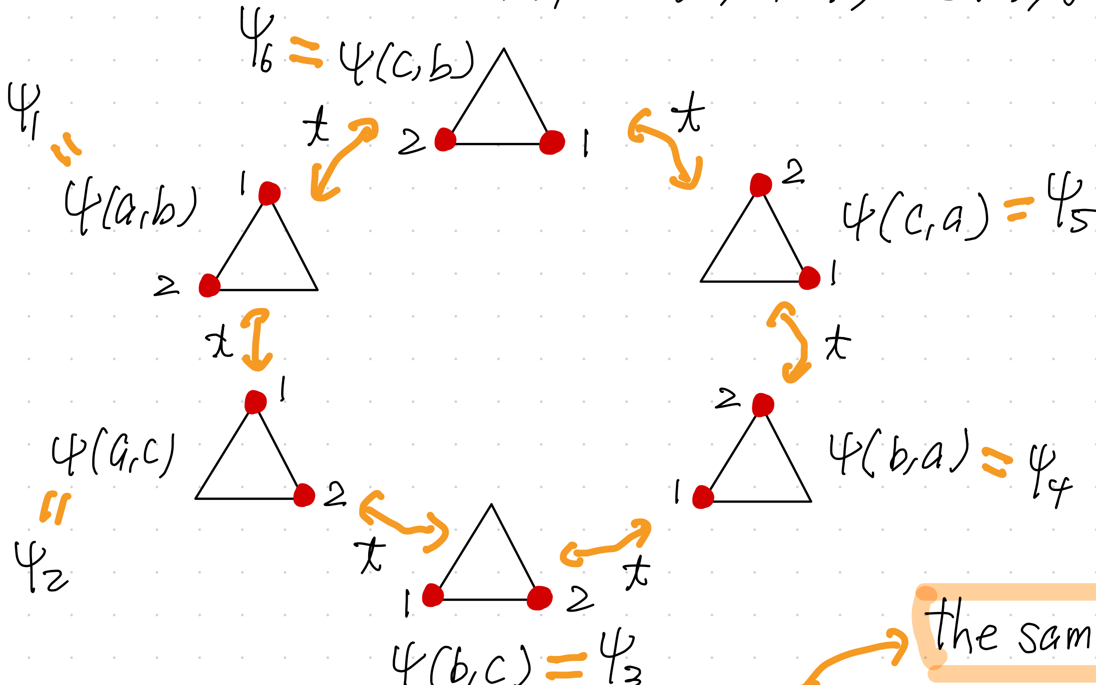
$$E\,\psi(a,b) = t\,\{\psi(c,b) + \psi(a,c)\} \quad (1)$$

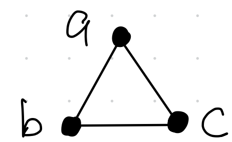


Schrödinger equation

$$E\psi(x,y) = t\{\psi(z,y)+\psi(x,z)\} \qquad (1)$$

$$x,y,z\in\{a,b,c\},\quad x\neq y,\ y\neq z,\ z\neq x \qquad (2)$$





The same as the simplest model in p.15!!

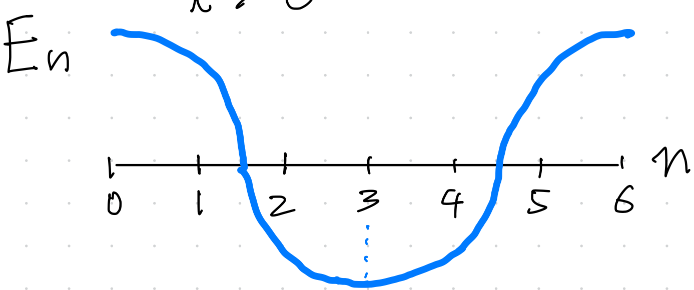
$$E\psi_j = t\{\psi_{j+1}+\psi_{j-1}\}, \qquad j=1,\cdots,6\ (\text{p.b.c}) \qquad (3)$$

energy eigenfunction $\psi_j^{(n)} = e^{i\frac{2\pi}{6}nj}$ (1)

energy eigenvalue $E_n = 2t\cos\left(\frac{2\pi}{6}n\right)$ (2)

where $n = 1, \cdots, 6$

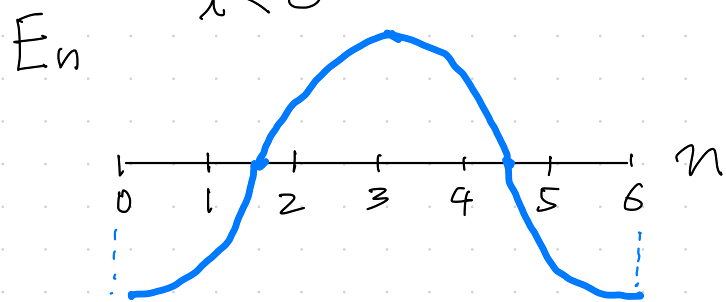
$t > 0$



the ground state $n = 3$

$\psi_j^{GS} = (-1)^j$ (3)

$t < 0$



the ground state $n = 0 = 6$

$\psi_j^{GS} = 1$ (4)

Symmetry and the total spin in the ground state

$$\psi_1 = \psi(a,b),\ \psi_2 = \psi(a,c),\ \psi_3 = \psi(b,c),\ \psi_4 = \psi(b,a),\ \psi_5 = \psi(c,a),\ \psi_6 = \psi(c,b) \quad (1)$$

- $t<0$ ground state $\psi_j = 1$ (2) $\psi(x,y) = \psi(y,x)$ (3) symmetric

  $\boxed{S_{tot}=0}$ antiferromagnetic coupling

  Wigner's theorem

- $t>0$ ground state $\psi_j = (-1)^j$ (4) $\psi(x,y) = -\psi(y,x)$ (5) antisymmetric

  $\boxed{S_{tot}=1}$ $\boxed{\text{ferromagnetic coupling!}}$

  Wigner's theorem?

the tight-binding model with $t>0$ cannot be an effective model of a two-electron system in continuum

$\boxed{\text{Heisenberg's insight!}}$

a better way to find the symmetry of the ground state

Schrödinger equation (p17-(3)) $E\Psi = T\Psi$ (1)

$\Psi = (\psi_j)_{j=1,\ldots,6}$ (2) $T = (t_{j,j'})_{j,j'=1,\ldots,6}$ (3) $t_{j,j'} = \begin{cases} t, & |j-j'|=1 \\ 0, & \text{otherwise} \end{cases}$ (4)

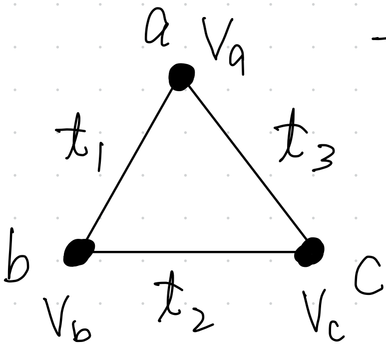
- $t < 0$ T satisfies the conditions for the Perron-Frobenius theorem
  the ground state is unique and can be chosen so that $\psi_j > 0$
  symmetry shows $\psi_j = \text{const.}$ symmetric

appendix 1

- $t > 0$ let $\tilde{\psi}_j = (-1)^j \psi_j$ then $E\tilde{\Psi} = -T\tilde{\Psi}$ (5)
  $-T$ satisfies the conditions for the Perron-Frobenius theorem
  the ground state is unique and can be chosen so that $\tilde{\psi}_j > 0$
  symmetry shows $\psi_j = (-1)^j \text{const.}$ antisymmetric

a generalization

bond-dependent hopping and site-depedent potential



the same two-electron problem

$t_1 t_2 t_3 < 0$

g.s. $\psi(x,y) = \psi(y,x)$

antiferromagnetic coupling

$t_1 t_2 t_3 > 0$

g.s. $\psi(x,y) = -\psi(x,y)$

ferromagnetic coupling

follows from the Perron-Frobenius Theorem